

# Socheat Chea

## AI Engineer · AI Training & Business Automation

Siem Reap, Cambodia    socheatsac7@gmail.com    (+855) 77 675 321  
linkedin.com/in/socheat-chea-6582a176

I build production software with AI coding agents, and I run it myself. Claude Code writes the code, Codex audits it independently, and I write the specs, set the guardrails, make the decisions and do every deploy. In 2026 this process put **five systems into production**, from a real-time crypto trading platform to a loyalty program used by a salon’s customers.

Before this I spent eight years in business roles: B2B sales for a corporate e-learning company, market research for clients around the world, digital marketing, and hotel online sales. I hold an MBA in International Business. That is why my AI work starts from a real business problem and ends with something people can use every day.

|                                        |                                                    |                                              |                                                 |
|----------------------------------------|----------------------------------------------------|----------------------------------------------|-------------------------------------------------|
| <b>5</b><br>systems live in production | <b>1,000+</b><br>commits across 7<br>product repos | <b>6,500+</b><br>automated test<br>functions | <b>200+</b><br>independent AI audit<br>verdicts |
|----------------------------------------|----------------------------------------------------|----------------------------------------------|-------------------------------------------------|

### HOW I WORK

## One AI builds, another AI audits, and I stay in charge

Every change starts as a written plan and passes two independent checks before it reaches real users.

- 1

**Write the card.** ME  
Scope, what must not change, and the bar it has to pass. Written before any code.
- 2

**Pre-build audit.** CODEX  
Reads the real code and the card, then returns PASS PASS WITH CONDITIONS or FAIL.
- 3

**Build.** CLAUDE CODE  
Implements the change switched off by default, adds tests, and hands back a diff with a report.
- 4

**Post-build audit.** CODEX  
Checks the diff and every claim in the report. A FAIL sends it back to step 3.

## 5 Deploy and watch. [ME](#)

I deploy and switch features on by hand, then check the live data.

### The auditor can't edit code.

Codex can only write to a folder outside every repository. It's a sandbox setting, not a request.

### The builder can't touch production.

Claude Code is blocked from pushing code, logging into servers, restarting services or reading secret files.

### Agreement isn't proof.

Two AIs agreeing is two opinions. Changes to live behaviour are proven with real forward data.

#### REAL EXAMPLES FROM THE REVIEW RECORD

##### **FAIL** Claude's tests mocked out the code under test

So a real crash never ran during testing. Its report also made a false claim. Fixed and passed in round 2.

##### **FAIL** A safety limit failed open after a restart

During a database outage the limit stopped working, even though all 69 tests were green. Fixed before release.

##### **FAIL** A payment QR code broke on one Python version

Caught because the tests run on two Python versions. Production used the one that broke.

More than half of first-round audits come back FAIL. That is the process working. Used on 3 projects with 130+ planned changes in total.

#### SELECTED WORK

## Five systems, all running today

### K+ crypto trading platform

● **Live, real money** Mar-Sep 2026 480 commits · ~111k lines · ~4,000 tests

Three services that stream live order data from four crypto exchanges (Binance, Bybit, OKX, Hyperliquid), score it with a rule engine, and place real trades on Bybit. A demo copy runs the same code, so every change is tested on demo before it goes live.

- **Data collector** for 7 live exchange feeds and 6 coins, updated every 5 seconds and served over a signed private API.
- **Signal engine** that checks every trade idea against a chain of rules. Restructured from one 3,858-line file into six modules, with tests proving nothing changed.
- **Trade executor** with safety first: a stop-loss must be confirmed within 5 seconds or the position is closed, plus a kill switch and constant checks against the exchange.
- **A/B test harness** that pairs demo and live trades and compares them with a proper statistical test.

## HARDEST PROBLEM

The data collectors were silently dropping trades. Five of six kept only the last trade in each batch from the exchange. I had counters added first to measure it: **65–81% of trading volume** on two exchanges was being lost. After the fix, all older data was marked unreliable and kept out of future decisions.

**Tech:** Python (asyncio), WebSockets, FastAPI, SQLite, pandas, Bybit API, PM2, cron, Telegram alerts

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## LinkedIn content agent

● **Live, weekly** Sep 2026, 10 days to launch 44 commits · 100 tests

An AI agent that drafts LinkedIn posts teaching Cambodian business leaders how to use AI in real work. It writes each post and its image, checks them, and sends them to me in Telegram. Nothing is published until I approve it. My LinkedIn posts about AI at work are drafted by this agent.

- **Structured output.** The model must return a strict JSON format, and every answer is validated before use.
- **Automatic quality checks.** Rules catch made-up stories, promised results and sales language. A failed draft goes back to the model with the exact problems, up to three tries.
- **Facts with sources.** Every number in a post is matched against a list of sourced facts and shown to me before I approve.
- **Contained AI.** The model runs locked down, with no internet tools and no access to publishing passwords.

**Tech:** Python, Pydantic, Codex CLI (first version on the Claude API), Telegram Bot API, SQLite, Typefully

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## Salon loyalty bot

ខ្មែរ · EN

● **Live with customers** Aug–Sep 2026, launched in 5 weeks 303 commits · 4,124 tests

A Telegram loyalty program that replaced paper stamp cards for a salon in Cambodia. Staff give points at the counter. Customers check their points, rewards and a personal QR code in Khmer or English.

- **A points record that can't be edited.** The database itself blocks changes and deletions, and corrections are added as new entries, like accounting.
- **A simple staff flow:** enter the amount, preview, confirm.
- **Khmer first.** All Khmer text is written and approved by the owner.
- **Tested hard.** Deliberate bugs are planted after each change to prove the tests catch them. The last two changes caught 77 of 77 and 34 of 34.

**Tech:** Python, aiogram, PostgreSQL, SQLAlchemy, Alembic, pytest, GitHub Actions

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## Ball Khmer football bot

ខ្មែរ · EN

● **Live** Feb–Aug 2026 201 commits · 845 tests

A Telegram bot for Cambodian football fans with live scores, fixtures, a prediction game, match alerts and a paid VIP tier, in Khmer and English.

- **Prediction game** with points, streaks, badges, a monthly leaderboard and referral codes.
- **Live alerts** every 60 seconds during matches, with no duplicates, and morning previews in each user's language.
- **Prediction tracking.** Match predictions come from an outside ML service. The bot saves each one before kickoff and the result after, so accuracy can be measured honestly over time.

### HARDEST PROBLEM

When a data provider failed, the bot remembered “no matches today” and sent that wrong news to every user in the morning. Now it says the data is unavailable instead of guessing, and the audit caught two more timing bugs before release.

**Tech:** Python, python-telegram-bot, SQLAlchemy, SQLite, Redis, APScheduler

## Crypto news brief

● **Live** Sep–Oct 2026 37 commits · 283 tests

A private Telegram news service: two daily digests plus 24/7 alerts on hacks, exchange outages and US regulation, collected from 11 free sources with a source attached to every item.

- **No AI where it isn't needed.** Simple rules handle selection today. An AI model will be added only when a test shows it improves the result enough to justify the cost.
- **Tested on real output.** When a review of live digests failed on quality, version 3 fixed it by merging duplicate reports of the same incident into one story.

**Tech:** Python, SQLite, systemd timers, Telegram

### HOW I TEST IDEAS

## Plan the test first, keep the “no” answers

Before changing a live system I write down the test in advance: what I expect, how I'll measure it, and what result would prove me wrong. Most answers were “no”, and keeping them stopped bad changes from going live.

| QUESTION                              | METHOD                                                   | RESULT          |
|---------------------------------------|----------------------------------------------------------|-----------------|
| Can smarter profit targets fix exits? | 121 rules tested on separate past and future data        | 0 of 121 passed |
| Is stop-loss placement the problem?   | Studied every winning and losing trade, replayed 7 rules | Stops were fine |
|                                       |                                                          |                 |

| QUESTION                                       | METHOD                                                              | RESULT           |
|------------------------------------------------|---------------------------------------------------------------------|------------------|
| Does the one rule that survived hold up live?  | Ran it on demo with an automatic kill switch, then A/B against live | <b>Retired</b>   |
| Do entries come too late after a market shift? | Statistical timing test on 162 signals                              | <b>No effect</b> |

## EXPERIENCE

### Business roles before building with AI

Jan 2022 – now

#### Founder

The Knowledge Plus · Cambodia

Crypto market education: daily videos, chart explanations and training. Since 2026, I've also built and run the K+ trading platform and the bots above.

Jan 2023 – now

#### Digital Specialist

Ti-Phsar · Cambodia

Websites, social media, paid ads and hotel yield management for clients.

Jan 2020 – Apr 2025

#### Freelance Market Researcher

Ravenry · Cambodia

Competitor research, data gathering, marketing analysis and content writing for clients around the world.

Nov 2019 – Nov 2020

#### Business Development Representative

Speexx · Thailand

B2B sales for a corporate e-learning company. Found decision-makers, set up meetings with company leaders, and handled negotiation using HubSpot, SalesLoft and LinkedIn Sales Navigator.

Jan 2018 – Sep 2019

#### Market Research Analyst (part-time and internship)

Canvassco · Thailand

Desk research, competitor and trend reports in English, and data analysis in Excel and PowerPoint.

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Jun 2017 – Nov 2018

## Social Media and Online Marketing Assistant Manager

Chan Angkor Boutique · Siem Reap

Online and seasonal promotions, and hotel booking channels (Expedia, Booking.com, Agoda, Airbnb).

### EDUCATION

## Degrees and training

### MBA, International Business

Panyapiwat Institute of Management, Thailand · 2020–2022

### BBA, International Business

Dhurakij Pundit University, Thailand · 2016–2019

### Information Technology training

University of South-East Asia · 2015–2016

### English for General Education

Australian Centre for Education · 2014–2016

### SKILLS

## What I work with

### AI coding agents

Claude Code, Codex CLI, project instructions (CLAUDE.md, AGENTS.md), custom skills, sandbox and permission setup, builder–auditor review loops

### LLM applications

Structured JSON output, automatic validation and retry, human approval steps, prompt caching, running models safely

### Backend

Python, FastAPI, Telegram bots (aiogram, python-telegram-bot), PostgreSQL, SQLite, Redis, SQLAlchemy, REST and WebSocket APIs

### Operations

Linux servers, PM2, systemd, cron, GitHub Actions, backups, monitoring and alerts

### Testing and data

pytest, mutation testing, A/B testing, statistics, pandas, Excel, market research

### Business

B2B sales and negotiation, digital marketing, customer relationships, staff training

### LANGUAGES

## I work in English

### English

Full professional

### Khmer (ខ្មែរ)

Native

### Thai

Elementary

### ABOUT THE CODE

My code repositories are private because they hold live trading systems and client data. I'm happy to walk you through any project on a screen share, including the plans, audit results, tests and deploy history. You can also try the Ball Khmer bot live in Telegram at @Ball\_khmerbot.